

Setup & user guide: know the carbon *before* you choose.

CarbonShift predicts the footprint of a decision (a trip, a purchase, a building project) while you can still change it, and shows the lower-carbon option. The AI reads your plan; every number comes from cited emission factors and live data.

1 • Set it up

1 Install the prerequisites

Python 3.11+ (python.org, tick "Add to PATH") and Node.js 20+ (nodejs.org).

2 Add your keys

The first run creates `backend\.env` and `frontend\.env.local`. Fill them in:

Required

- `GROQ_API_KEY` in backend: AI understanding (free key at console.groq.com)
- `SUPABASE_URL` + `SUPABASE_SERVICE_ROLE_KEY` in backend
- `NEXT_PUBLIC_SUPABASE_URL` + `NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY` in frontend
- Optional more accurate data; the built-in data is used without them
- `OPENROUTESERVICE_API_KEY` : real road routes
- `GOOGLE_TRAVEL_IMPACT_API_KEY` : flight emissions
- `EMBER_API_KEY` , `ELECTRICITYMAPS_API_KEY` : grid carbon, cleanest hour
- `NEXT_PUBLIC_MAPTILER_KEY` : nicer map tiles

3 Prepare Supabase (one time)

Create a project → SQL Editor → run `backend/supabase_schema.sql`. Under Authentication → URL Configuration, set the site URL to `http://localhost:3000`.

4 Run everything with one command

From the project folder (or double-click `run.cmd`):

```
.\run.cmd
```

The first run installs everything. The app opens at `localhost:3000` and the API runs on port 8000. To stop it, press `Ctrl+C` or run `stop.cmd`.

If something goes wrong

- "Account setup required": the Supabase keys in `frontend\.env.local` are missing. Add them and run again.
- Predictions fail: check `GROQ_API_KEY`; the free tier has a daily token limit.
- Port already in use: `run.cmd` stops an old copy itself; otherwise run `stop.cmd`.
- Never share or commit the two `.env` files.

2 • Use it

1 Create an account

Go to **Create account** with your email and a password. Your history, choices and goals sync to your account.

2 Predict a decision

Under **Predict a decision**, describe the plan in plain words or with the mic, e.g. *"Going from Mohali to Delhi"* or *"Building a 2 storey house in Mohali"*. A distance is optional. You get:

- kg CO₂e with Scope 1/2/3 and a cited breakdown
- greener alternatives with savings and rough cost
- for trips: every travel mode compared on a route map

Ask follow-up questions in the chat to refine it. More detail gives a more accurate result.

3 Record what you actually did

Pick the option (or tick several changes), then confirm it once you've made the switch:

Pick option → Planned → **✓ I did it** → Counted as avoided

The dashboard's **"Did you go greener?"** card reminds you about planned changes.

4 Plan with others

Meeting point finds the lowest-carbon venue for a group travelling from different cities. **Calculator** gives an exact figure when you know the activity and amount.

5 Track your progress

Dashboard: totals, charts, monthly budget and filters. **History**: every decision with its recommendations. **Recommendations**: your best changes ranked.

6 Plan ahead and report

Ghost You shows the future where you always take the best option. **Net-zero plan** ranks changes by cost per tonne towards your target year. **Report** prints all, filtered or single decisions as a PDF.